

UNIVERSITY OF BARISHAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SOFTWARE ENGINEERING PROJECT PROPOSAL

GroupSync – Intelligent Group Project Management with Contribution Analytics

CSE-3104: Software Engineering and Information System Design Lab

Submitted By:

Team No.: 03

Team Name: SlackerSlayer

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Formation of Team SlackerSlayer

Our project team consists of six dedicated members working collaboratively to design, develop, test, and deploy the GroupSync platform. The team structure has been organized carefully to ensure efficient task distribution and smooth project management throughout the development cycle.

Team Composition and Responsibilities

Member 1: Backend & Project Lead (Team Leader)

Name: Humayra Nuha **Responsibilities:**

- Lead overall project planning and sprint management.
- Develop backend services using Node.js and Express.js.
- Design authentication and authorization systems.
- Manage deployment and API integration.
- Coordinate collaboration among all members.

Member 2: Database Designer & Requirement Analyst

Name: Puja Gain

Responsibilities:

- Gather and analyze system requirements.
- Design MongoDB database schema and relationships.
- Create ER diagrams and architecture models.
- Ensure data consistency and optimization.
- Support backend integration.

Member 3: Frontend Lead & UI/UX Designer

Name: Moumita Ghose **Responsibilities:**

- Develop responsive frontend interfaces using Next.js and TailwindCSS.
- Design dashboards for students and professors.
- Create wireframes and UI prototypes.
- Ensure frontend-backend integration consistency.
- Improve overall user experience.

Member 4: Frontend Developer & Documentation Officer

Name: Ritu Akter Samia **Responsibilities:**

- Assist in frontend component development.
- Maintain documentation and reports.
- Prepare presentation slides and user manuals.
- Support GitHub version control management.
- Coordinate project communication.

Member 5: Tester & Backend Developer

Name: Sanjida Akter Mim **Responsibilities:**

- Perform unit and integration testing.
- Design test cases for APIs and modules.
- Assist backend development and debugging.
- Verify contribution scoring accuracy.

- Conduct security and performance testing.

Member 6: Maintenance & Deployment Engineer

Name: Maimuna Akter **Responsibilities:**

- Manage deployment and server monitoring.
- Maintain CI/CD pipelines and hosting.
- Handle bug fixes and optimization.
- Monitor system logs and security updates.
- Ensure long-term system stability.

Note: All team members will participate in frontend and backend activities during Agile sprints to encourage collaborative learning and balanced workload distribution. GitHub will be used for version control and Trello/Slack for project coordination.

1.1 SWOT Analysis of Team SlackerSlayer

Strengths

1. Strong understanding of the MERN stack and full-stack web development.
2. Good teamwork and role distribution using Agile methodology.
3. Experience with GitHub collaboration and version control.
4. Knowledge of responsive UI design and REST API integration.
5. Ability to quickly learn new technologies and tools.

Weaknesses

1. Limited experience with large-scale real-time systems.
2. Dependency on third-party APIs such as GitHub and OpenAI.
3. Academic workload may affect sprint timelines.
4. Limited budget for premium cloud services.

Opportunities

1. Solve a real-world academic problem faced by universities.
2. Gain practical experience in full software development lifecycle.
3. Expand the platform for industry team collaboration in future versions.
4. Learn AI-assisted analytics and real-time system integration.

Threats

1. GitHub API limitations or service interruptions.
2. Incorrect peer reviews affecting scoring fairness.
3. Potential security vulnerabilities if authentication is weak.
4. Hosting downtime during demonstrations or testing.

Overview

GroupSync is an intelligent group project management platform designed to solve the “free rider” problem commonly found in academic group assignments. In many university projects, a small number of students complete most of the work while others contribute very little but still receive equal grades. Professors often lack objective evidence to fairly evaluate each student’s contribution.

GroupSync addresses this issue by combining GitHub integration, peer reviews, attendance tracking, and contribution analytics into one centralized platform. The system automatically collects development activity data and generates individual contribution scores using a weighted algorithm.

The platform provides separate dashboards for students, professors, and administrators. Students can manage groups, connect GitHub repositories, log work hours, and submit peer reviews. Professors can monitor group performance, detect free riders, and generate detailed reports for fair grading decisions.

Additionally, the platform uses lightweight AI analysis to assist in detecting suspicious contribution patterns and identifying students at risk of becoming free riders.

Overall, GroupSync modernizes academic project management by improving transparency, accountability, collaboration, and fairness within university group work.

Problem Statement

Group projects are an important part of university education because they encourage collaboration, communication, and teamwork skills. However, traditional group project evaluation methods suffer from several major problems.

In most cases, professors cannot accurately track the contribution of each student. A few students often complete the majority of the work while others contribute very little. Despite this imbalance, all members usually receive the same grade, creating frustration among hardworking students.

Existing tools such as GitHub Insights only track commits and fail to capture other important aspects such as attendance, communication, teamwork, and peer feedback. Manual peer evaluation forms are time-consuming and often unreliable. Self-reported working hours may also be inaccurate or dishonest.

As a result, professors lack objective data to justify individual grading decisions. This creates unfair evaluation systems and reduces accountability in collaborative learning environments.

Therefore, there is a need for a centralized intelligent platform that combines automated contribution tracking, peer reviews, attendance monitoring, and AI-assisted analysis to generate fair and transparent contribution reports for academic group projects.

Motivation

The motivation behind GroupSync comes from the common challenges students face during university group projects. Many students experience situations where only a few team members actively work while others remain inactive yet receive equal recognition.

As computer science students, we wanted to develop a solution that promotes fairness, accountability, and transparency in collaborative academic work. We were inspired by modern project management systems used in software industries and aimed to adapt similar ideas for educational environments.

The project also provides an opportunity to learn advanced software engineering concepts such as:

- Full-stack web development
- Real-time communication
- API integration
- Authentication and security
- Contribution analytics
- AI-assisted decision support

GroupSync not only solves a practical academic problem but also helps us gain valuable real-world development experience.

User & Admin Requirements

Students

Students are the primary users of the system and interact with the platform to manage academic group projects.

Registration & Authentication

- Register using university email.
- Secure login using JWT authentication.
- Connect GitHub account through OAuth.
- Manage personal profile information.

Group Management

- Create groups and generate invite codes.
- Join groups using invite links or codes.
- View team members and project details.
- Leave or switch groups when permitted.

Contribution Tracking

- Connect project repositories from GitHub.
- Automatically track commits and code contributions.
- Log working hours and attendance.
- Receive real-time contribution score updates.

Peer Reviews & Analytics

- Submit anonymous peer reviews.
- Rate teammates based on teamwork and communication.
- View personal contribution analytics.
- Receive warnings for low contribution scores.

Professors

Professors supervise courses and evaluate student performance.

Course & Group Management

- Create courses and assign students.
- Monitor all groups within a course.
- View repository activity and contribution records.

Monitoring & Evaluation

- Detect free riders automatically.
- Monitor peer review summaries.
- Analyze weekly contribution reports.
- Compare student performance fairly.

Reports & Analytics

- Generate PDF reports.
- Export analytics to CSV.
- View charts and visual dashboards.
- Access AI-generated risk analysis.

Admin Panel

Responsibilities

- Manage users and roles.
- Monitor platform activity.
- Configure system settings.
- Handle security and maintenance tasks.

Security & Monitoring

- Maintain audit logs.
- Monitor suspicious activities.
- Ensure secure API communication.
- Manage backups and recovery systems.

Development Approach

SDLC Approach — Agile Methodology

The project will follow the Agile Development Model with iterative development cycles and weekly sprint planning.

Advantages:

- Faster incremental development
- Continuous testing and feedback
- Flexible requirement handling
- Better team collaboration

Tools & Technology

Frontend

- Next.js 14
- React.js
- TypeScript
- TailwindCSS
- Socket.io Client

Backend

- Node.js
- Express.js
- JWT Authentication
- bcrypt
- GitHub OAuth API

Database

- MongoDB Atlas
- Mongoose ODM

Security Measures

- JWT-based authentication
- Password hashing with bcrypt
- Role-based access control
- HTTPS communication

Deployment & DevOps

- Vercel for frontend deployment
- Railway for backend hosting

- GitHub for version control
- MongoDB Atlas cloud hosting

Project Timeline (8 Weeks Plan)

Week	Deliverables
1	Requirement analysis, architecture planning
2	Authentication system and database setup
3	Group management and CRUD operations
4	GitHub OAuth and webhook integration
5	Contribution scoring engine
6	Peer review system and dashboards
7	AI analysis and report generation
8	Testing, deployment, and documentation

Software Development Life Cycle

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

Database Design & Contribution Algorithm

The system will use six major MongoDB collections:

- Users
- Courses
- Groups

- Contributions
- PeerReviews
- GitHubEvents

The contribution scoring system combines:

- Commit count
- Lines of code added
- Messages and communication
- Attendance
- Logged work hours
- Peer review ratings

The final contribution score is calculated on a scale from 0–100 using weighted metrics and penalty rules.

AI Integration

The platform uses lightweight AI analysis to enhance free rider detection.

AI Features:

- Analyze low-performing students
- Detect suspicious contribution behavior
- Generate recommendation reports
- Provide confidence-based evaluation

The AI module only triggers for students with low contribution scores to reduce API costs and improve efficiency.

Budget and Estimation

		Estimated
Category	Description	Cost (BDT)
Development & Team Compensation	Salaries and allowances for developers, designers, testers, and coordinators during the project development cycle	850,000
UI/UX Design & Prototyping	Figma Pro, design assets, prototyping, and user experience testing	120,000

Software Tools & Subscriptions	GitHub Pro, Postman, API tools, collaboration software, and project management tools	90,000
Cloud Hosting & Deployment	Vercel, Railway, domain registration, SSL certificate, CDN, and cloud deployment services	110,000
Database & Storage Services	MongoDB Atlas cloud database, backups, and storage services	80,000
AI Integration & API Costs	OpenAI API usage, analytics processing, and AI testing expenses	140,000
Security & Testing	Security audits, penetration testing, performance testing, and QA operations	130,000
Internet & Maintenance	Internet, electricity backup, server maintenance, and technical support	70,000
Documentation & Printing	Project proposal, reports, presentation materials, and printing costs	25,000
Research & Expert Consultation	Consultation for software architecture, AI integration, and system security review	160,000
Marketing & User Awareness	Promotional materials, demo presentation, banners, and awareness campaigns	75,000
Contingency & Future Maintenance	Emergency funds, bug fixing, future updates, and postdeployment support	150,000

|| Total Estimated Budget | 2,000,000 BDT |

Conclusion

GroupSync is a modern intelligent project management platform designed to improve fairness and accountability in academic group work. By integrating GitHub analytics, peer reviews, attendance tracking, and AI-assisted contribution analysis, the system provides professors with objective evidence for evaluating student performance.

Using Agile methodology and the MERN technology stack, the project will be developed as a scalable and secure full-stack web application. The platform reduces manual monitoring effort, encourages active participation, and helps solve the long-standing free rider problem in university group projects.

Overall, GroupSync represents an innovative and practical software engineering solution that combines collaboration, analytics, and intelligent automation to improve academic project management systems.

